Cuckoo Search

Pseudo code

Begin
initialize n host nests randomly
Evaluate fitness of all nests
while stopping condition is not met
Generate a new solution using Lévy flight
Evaluate its fitness
Select a random nest j
if new solution is better than nest j
Replace nest j with new solution
End if
Abandon a fraction pa of the worst nests
Generate new nests randomly
Keep the best solutions
End while
Return best solutions
End

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Start
input objective function, n, Pa, max iteration
Generate n random nests
Evaluate fitness of all nests
Find best fitness
for i = 1 to max_iterations
    for each nest i
        new_nest = nest[i] + Levy_flight(nest[i])
        Evaluate fitness (new_nest)
        if fitness (new_nest) is better
            nest[i] = new_nest
        End if
    End for

Abandon Pa fraction of worst nest
Replace abandoned nests with random sol
update best_nest
End for

Output best_nest & its fitness
Stop

Levy_flight (nest)
    generate random step using Levy dist
    new_nest = nest + step
    return new_nest

Mechanical = Design

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Cuckoo Search via Lévy flights (CS)

It introduces cuckoo search, a nature inspired optimization algorithm based on cuckoo breeding behaviour & Lévy flights. The algorithm represents candidate solutions as nests & retains better solutions over successive iterations. Some nests are abandoned to encourage exploration. The authors evaluate the algo using benchmark optimization problems & compare its performance with other optimization methods. CS maintains to balance exploration & exploitation. It can be adapted to various engineering optimization problems. The paper provides the theoretical foundation for subsequent CS-based mechanical design studies.

Code Summary

The python implementation applies CS to minimize the mass of a solid circular shaft. It defines the shaft's material properties, applied torque, length & design constraints. Shaft mass is used as the objective function, while shear stress & angle of twist are calculated to check feasibility. The program initializes 25 candidate shaft diameters within specified bounds. Lévy flights generate new candidate diameters during each iteration. Better solutions replace inferior candidates & some nests are abandoned for encouragement. The algo runs for 500 iterations & tracks best feasible design.

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Assignment Summary

For the shaft design problem, the analytical reference diameter is approximately 42.23 mm corresponding to a mass of about 11.0 kg. The associated shear stress is approximately 21.1 MPa, below the allowable limit of 80 MPa. The angle of twist is approximately 0.020 rad, which meets the specified maximum limit. Therefore, the twist constraint determines the minimum feasible diameter.

Checked

COR
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